Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Выполнить приимеры

2 Выполнить дествия по работе с каталогами и файлами

3 Выполнить действия с правами доступа

4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы

```
aaharlamova@aaharlamova:~$ touch abc1
aaharlamova@aaharlamova:~$ cp abc1 april
aaharlamova@aaharlamova:~$ cp abc1 may
aaharlamova@aaharlamova:~$ mkdir monthly
aaharlamova@aaharlamova:~$ cp april may monthly
aaharlamova@aaharlamova:~$ cp monthly/may monthly/june
aaharlamova@aaharlamova:~$ ls monthly
april  june  may
aaharlamova@aaharlamova:~$ mkdir monthly.00
aaharlamova@aaharlamova:~$ cp -r monthly monthly.00
aaharlamova@aaharlamova:~$ cp -r monthly.00 /tmp
aaharlamova@aaharlamova:~$ 
```

Рисунок 1: Выполнение примеров

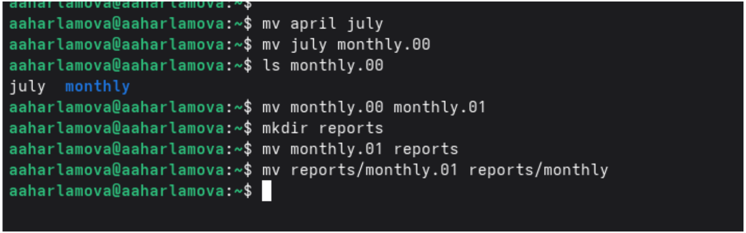


Рисунок 2: Выполнение примеров

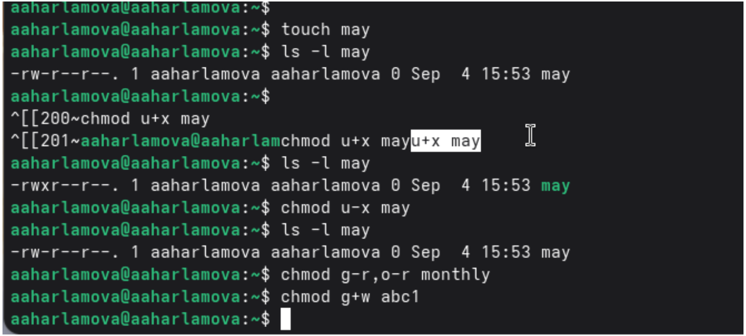


Рисунок 3: Выполнение примеров

Создание директорий и копирование файлов

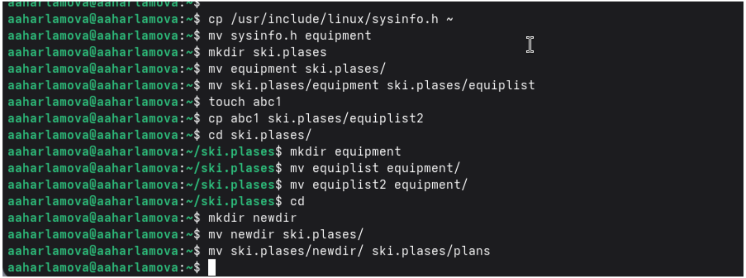
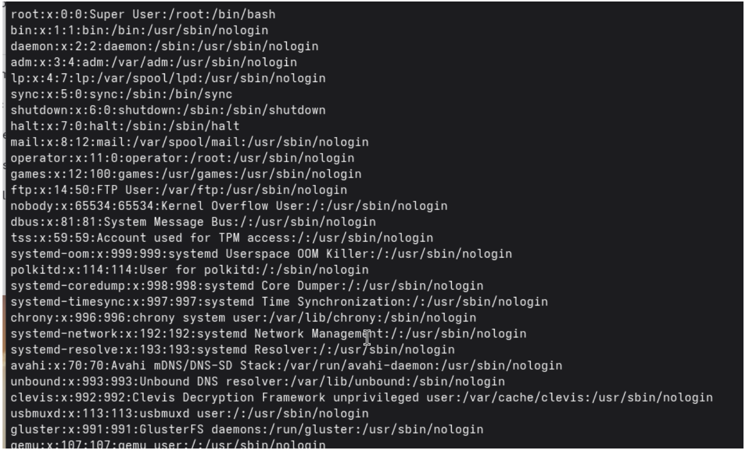


Рисунок 4: Работа с каталогами

Работа с командой chmod

```
aaharlamova@aaharlamova:~$ mkdir australia play
aaharlamova@aaharlamova:~$ touch my_os feathers
aaharlamova@aaharlamova:~$ chmod 744 australia/
aaharlamova@aaharlamova:~$ chmod 711 play/
aaharlamova@aaharlamova:~$ chmod 544 my_os
aaharlamova@aaharlamova:~$ chmod 664 feathers
aaharlamova@aaharlamova:~$ ls -l
total 20
-rw-rw-r--. 1 aaharlamova aaharlamova     0 Sep  4 15:54 abc1
drwxr--r--. 1 aaharlamova aaharlamova     0 Sep  4 15:56 australia
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Desktop
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Documents
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Downloads
-rw-rw-r--. 1 aaharlamova aaharlamova     0 Sep  4 15:56 feathers
drwxr-xr-x. 1 aaharlamova aaharlamova    74 Sep  4 15:27 git-extended
-rw-r--r--. 1 aaharlamova aaharlamova     0 Sep  4 15:53 may
drwx--x--x. 1 aaharlamova aaharlamova    24 Sep  4 15:51 monthly
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Music
-r-xr--r--. 1 aaharlamova aaharlamova     0 Sep  4 15:56 my_os
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Pictures
drwx--x--x. 1 aaharlamova aaharlamova     0 Sep  4 15:56 play
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Public
drwxr-xr-x. 1 aaharlamova aaharlamova    14 Sep  4 15:51 reports
drwxr-xr-x. 1 aaharlamova aaharlamova    28 Sep  4 15:54 ski.plases
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Templates
-rw-r--r--. 1 aaharlamova aaharlamova 16555 Sep  4 15:44 tory
drwxr-xr-x. 1 aaharlamova aaharlamova     0 Sep  4 14:50 Videos
drwxr-xr-x. 1 aaharlamova aaharlamova    10 Sep  4 15:01 work
aaharlamova@aaharlamova:~$
```

Рисунок 5: Настройка прав доступа



```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:/usr/sbin/nologin
```

Рисунок 6: Файл /etc/passwd

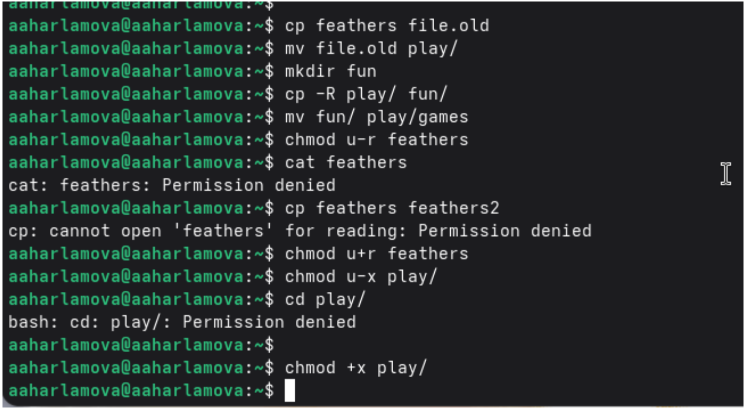
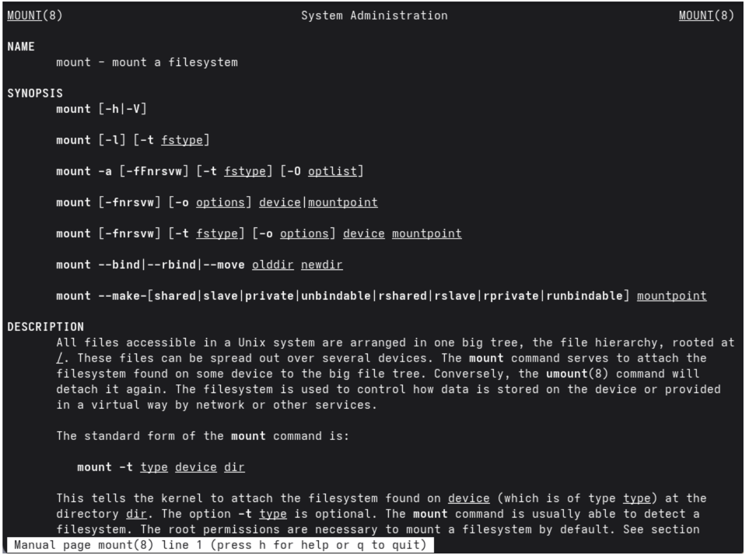


Рисунок 7: Работа с файлами и правами доступа



```
MOUNT(8)                      System Administration                      MOUNT(8)

NAME
       mount - mount a filesystem

SYNOPSIS
       mount [-h|-V]

       mount [-l] [-t fstype]

       mount -a [-fFnrsvw] [-t fstype] [-O optlist]

       mount [-fnrsvw] [-o options] device|mountpoint

       mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

       mount --bind|--rbind|--move olddir newdir

       mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION
       All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at
       /. These files can be spread out over several devices. The mount command serves to attach the
       filesystem found on some device to the big file tree. Conversely, the umount(8) command will
       detach it again. The filesystem is used to control how data is stored on the device or provided
       in a virtual way by network or other services.

       The standard form of the mount command is:

          mount -t type device dir

       This tells the kernel to attach the filesystem found on device (which is of type type) at the
       directory dir. The option -t type is optional. The mount command is usually able to detect a
       filesystem. The root permissions are necessary to mount a filesystem by default. See section
 Manual page mount(8) line 1 (press h for help or q to quit)
```

Рисунок 8: Команда mount

Справка по командам

```
FSCK(8)                      System Administration                      FSCK(8)

NAME
       fsck - check and repair a Linux filesystem

SYNOPSIS
       fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
       fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a
       device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem
       label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root).
       Normally, the fsck program will try to handle filesystems on different physical disk drives in
       parallel to reduce the total amount of time needed to check all of them.

       If no filesystems are specified on the command line, and the -A option is not specified, fsck
       will default to checking filesystems in /etc/fstab serially. This is equivalent to the -As
       options.

       The exit status returned by fsck is the sum of the following conditions:

       0
           No errors

       1
           Filesystem errors corrected

       2
           System should be rebooted

       4
           Filesystem errors left uncorrected

       8
 Manual page fsck(8) line 1 (press h for help or q to quit)
```

Рисунок 9: Команда fsck

Справка по командам

```
MKFS(8)                         System Administration                         MKFS(8)

NAME
       mkfs - build a Linux filesystem

SYNOPSIS
       mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
       This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

       mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device
       argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall
       contain the filesystem. The size argument is the number of blocks to be used for the filesystem.

       The exit status returned by mkfs is 0 on success and 1 on failure.

       In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype)
       available under Linux. The filesystem-specific builder is searched for via your PATH environment
       setting only. Please see the filesystem-specific builder manual pages for further details.

OPTIONS
       -t, --type type
           Specify the type of filesystem to be built. If not specified, the default filesystem type
           (currently ext2) is used.

       fs-options
           Filesystem-specific options to be passed to the real filesystem builder.

       -V, --verbose
           Produce verbose output, including all filesystem-specific commands that are executed.
           Specifying this option more than once inhibits execution of any filesystem-specific commands.
           This is really only useful for testing.

 Manual page mkfs(8) line 1 (press h for help or q to quit)
```

Рисунок 10: Команда mkfs

```
KILL(1)                              User Commands                              KILL(1)

NAME
       kill - terminate a process

SYNOPSIS
       kill  [-signal|-s signal|-p]  [-q value] [-a] [--timeout milliseconds signal] [--] pid|name...

       kill -l [number|0xsigmask] | -L

       kill -d pid

DESCRIPTION
       The command kill sends the specified signal to the specified processes or process groups.

       If no signal is specified, the TERM signal is sent. The default action for this signal is to
       terminate the process. This signal should be used in preference to the KILL signal (number 9),
       since a process may install a handler for the TERM signal in order to perform clean-up steps
       before terminating in an orderly fashion. If a process does not terminate after a TERM signal has
       been sent, then the KILL signal may be used; be aware that the latter signal cannot be caught,
       and so does not give the target process the opportunity to perform any clean-up before
       terminating.

       Most modern shells have a builtin kill command, with a usage rather similar to that of the
       command described here. The --all, --pid, and --queue options, and the possibility to specify
       processes by command name, are local extensions.

       If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
       The list of processes to be signaled can be a mixture of names and PIDs.

       pid
           Each pid can be expressed in one of the following ways:
 Manual page kill(1) line 1 (press h for help or q to quit)
```

3. Выводы по проделанной работе

Вывод

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.